

Regression Algorithm Assignment

Clients Requirement:

1.Problem statement:

Client wants to predict the insurance charges based on several parameters. Client has provided the dataset.

Need to develop a model which predicts insurance charges.

2. The given dataset has parameters such as age, sex, BMI, children, smoker and charges.

When we check on the given dataset we can say the input parameters as age,sex, BMI, Children, smoker.

And Output as the charges.

Since we know the input and output parameters in the given dataset then

Stage1: **Domain – Machine learning** since the given data set having numbers.

Stage2: Type of learning: **Supervised learning** since we know the dataset with known input and output parameters.

Stage 3: In the data we have numerical values so under supervised learning we can say it as **Regression Algorithm**.

In the given dataset we have 1338 rows × 6 columns

For the column sex and smoker we need to convert the word to numerical value.

Tabulation for R-Score value

Linear Regression

Linear Regression	R_score
Simple linear regression	0.78

Multiple Linear Regression gives the same result as 0.78

SVM

Parameters	C-value	R-score
Linear	C=1000	0.63
Linear	C=100	0.54
Poly	C=100	-0.09
Poly	C=100	-0.05
rbf	C=1000	-0.11

Rbf	C=100	-0.12
<i>sigmoid</i>	C=100	-0.11
<i>sigmoid</i>	C=1000	-1.66
precomputed	C=1000	Given input should be square matrix.

Decision Tree Regressor

Parameters	Splitter	R-score
Poisson	Random	0.74
Poisson	Best	0.71
<i>squared_error</i>	Best	0.68
<i>friedman_mse</i>	Best	0.70
<i>absolute_error</i>	Best	0.67
<i>absolute_error</i>	random	0.71
<i>friedman_mse</i>	random	0.70
<i>squared_error</i>	Random	0.73

Random Forest

n-estimators	Random_state	R-score
50	0	0.84
100	0	0.85
10	0	0.83

The model from Random forest with n-estimators =100 and randomstate=0 gives the R-score 0.85 is the good model.